

A Graph-Theoretic Investigation of Brain Network Structure in Parkinson's Disease Using Resting-State fMRIs

Sean Meehan

Abstract

Connectomics is an emerging field that applies graph theory to study the structural and functional organization of the brain. This paper investigates macroscale brain network properties based on resting-state functional MRI data from the Neurocon dataset, which includes a set of control subjects and a set of individuals with Parkinson's disease. Correlation matrices were generated based on blood-oxygen-level-dependent signals between regions of interest in the brain using automated anatomical labeling parcellation. These matrices were then converted to adjacency matrices via binarization and global thresholding. These thresholded matrices were used to construct brain graphs, which were analyzed for key network statistics and small-world characteristics. 'Small-worldness' was quantified using a small-world coefficient, comparing real brain graphs to matched Erdos-Renyi random graphs. Results showed notable differences between the small-world-coefficients of the patient and control groups, and in graph density and node degree, suggesting underlying structural variation. These findings indicate that macro-level 'small-worldness' and, more broadly, underlying network structure and efficiency may be subtly affected in Parkinson's disease, warranting further investigation into blood-oxygen-level-dependent signal distributions and thresholding effects in functional brain networks.

Introduction

Networks have become ubiquitous in modern life. We drive on a network of roads; we constantly interact with technological networks; we share our lives on social networks. As a result of the predominance of networks, network science has become one of the predominant fields of scholarship. Network science, like the networks it seeks to understand, can exist in many different contexts and disciplines. One of these

contexts, our context, concerns the networks that comprise the brain. The discipline that arises from the use of network theory to study the brain is known as connectomics.

Put more precisely, connectomics is the “use of graph theory to model, estimate, and simulate the topology and dynamics of brain networks.” This is a relatively new field that is developing out of the recent, rapid “growth in the science of networks in general” and the “technological evolution of methods to measure and visualize brain organization” [1]. Connectomics seeks to represent the brain as a set of nodes and edges in order to determine what makes the brain such an effective network.

Central to connectomics is the idea of a small world graph. Graph representations of the brain are commonly found to be small world. The concept of a small world graph relies on the following two definitions.

The **clustering coefficient** (CC) is, given that $\{a,b\}, \{b,c\} \in E$, the probability that $\{a,c\} \in E$, where E is the set of edges of the graph G , and V is the set of vertices of G with $a,b,c \in V$. The **characteristic path length** (CPL) is the average minimum number of edges traversed to get from a given vertex to any other vertex in the graph.

The idea of small world graphs was popularized by Duncan Watts and Steven Strogatz as a graph with a low CPL and a high CC. They outlined a method to generate small world graphs by starting with a lattice graph and iteratively applying a probability, p , to each edge, $\{i,j\}$, that it would be changed to $\{i,k\}$, where $k \neq j$ is randomly selected from the rest of the vertices [2]. The effects on the CC and CPL as randomness increases can best be demonstrated by figures 1 and 2.

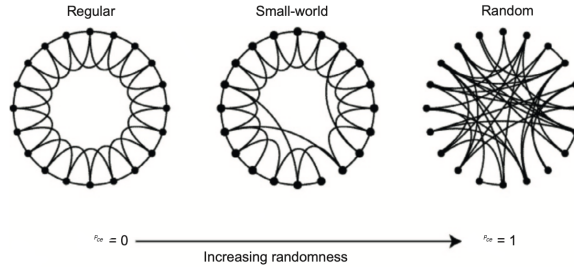


Figure 1: Effects on lattice graph as p increases. From [1]

We expect that setting a low value for p would not greatly decrease the value of the CC because the graph would still closely resemble a lattice. Simultaneously, we expect to see a decrease in the CPL because the few randomly changed edges act as bridges that connect areas of the graph that would otherwise be geodesically distant, as seen in the middle graph of figure 1. On the other hand, setting a higher value for p would cause both the CC and CPL to approach 0. Figure 2 shows the experimental results of Watts and Strogatz which confirm that this is indeed true [2].

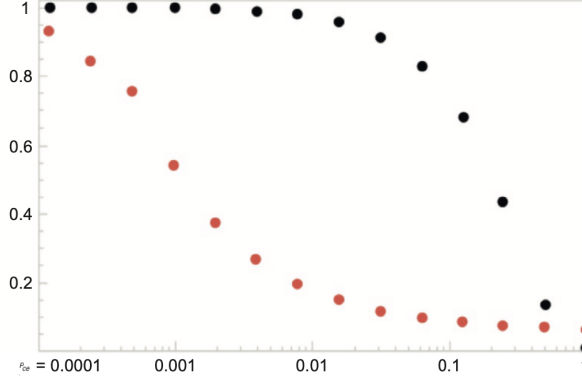


Figure 2: CC vs p in black and CPL vs p in red. From [1]

What we see is essentially a ‘sweet spot’ where the CC is high and the CPL is low. Graphs that exist in this sweet spot are called small world. A method for quantifying the small world characteristics of a given graph is given in the analysis section.

Another key concept in connectomics is the scale of the graph that is being studied. There are considered to be three scales on which connectomics can be used to study the brain. The first of these is the microscale. Connectomic consideration of the brain at the microscale involves graphs representing the brain’s physical, cellular structure, with neurons as nodes and synapses as edges. It is naturally difficult to develop an accurate graph representation of the human brain’s entire cellular structure, but the brains of smaller organisms have been successfully mapped including “the reconstruction of the nervous system of the nematode worm *C. elegans* at the level of each and every neuron and synapse”[1]. The second scale in connectomics is the mesoscale. Mesoscale graphs attempt to map “connectivity between neuronal populations or cell assemblies, rather than individual neurons”[1], using methods like tract tracing. The third and final scale is the macroscale. Macroscale connectomics involves constructing graphs to represent regions of the brain based off of scans like EEGs or MRIs. These techniques have much lower resolution and cannot give any information on the cellular level. However, the noninvasive nature of the data collection processes makes macroscale research the most common in connectomics.

Our study works on the macroscale with functional, time sequence correlation MRI data published in supplement to “Data-Driven Network Neuroscience: On Data Collection and Benchmark”[3]. The analysis of these data was performed using Python, with a reliance on the NetworkX [4] module for graph-theoretical analysis. All of the code used throughout this project is published on GitHub at this address: <https://github.com/SeanDMeehan/BrainNetworkResearch>.

This paper will be organized into the following sections.

1. Introduction

2. Data Preprocessing
3. Thresholding
4. Graph construction
5. Analysis
6. Results
7. Discussion
8. Conclusion

Data Preprocessing

The data being used for this project comes from [3]. As such, the first stages of the data preprocessing were performed by the authors of that paper. Those steps will be discussed here in brief. For a more in depth discussion, see [3].

The data used in [3] comes from the following studies or data sets: Autism Brain Imaging Data Exchange (ABIDE), Alzheimer’s Disease Neuroimaging Initiative (ADNI), Parkinson’s Progression Markers Initiative (PPMI), Matai, the TaoWu dataset, and the Neurocon dataset. All of these contain collections of raw resting state, functional magnetic resonance images (rs-fMRI) of varying numbers of patients.

These raw rs-fMRIs were converted into the Brain Imaging Data Structure (BIDS) format, a conventional storage system for such data. A certain amount of data processing and cleaning was then performed using a software called fMRIPrep to fully establish blood-oxygen-level-dependent (BOLD) signal images. The underlying principle here is that, as an area of the brain becomes more active, more blood flows into that area. The additional blood brings additional oxygen. The extra oxygen causes changes in the magnetic principles of this region of the brain that are measurable by fMRI. These measurements are BOLD signals, and images of the brain can be created from them.

After establishing the BOLD signal images, the data was parcellated into separate regions of interest (ROIs) using several different methods. These methods are separated into two categories. The first are atlas-based parcellation methods, which separate the brain into pre-defined ROIs. These methods were automated anatomical labeling (AAL), HarvardOxford, and Schaefer parcellation. The other two parcellation methods were clustering-based: k-means and WARD clustering. Once the data was separated into ROIs, a time sequence correlation matrix was generated by calculating the correlation over time between the BOLD signals of each individual ROI with all the others [3].

The data used in our study is from the Neurocon dataset, parcellated using the AAL method. The Neurocon dataset contains the rs-fMRIs of a control group and a patient group of individuals with Parkinson’s disease. The correlation matrices were downloaded from the University of Aucklands’s website [5]. Each matrix was in a .mat file. This project was done in Python, so our first step was to convert these into a compatible file type. A data processing module, DataProcessing, was written to convert these from .mat to .json and then from .json to .csv.

Thresholding

Recall that the data are correlation matrices of the time sequence correlations of BOLD signals between ROIs in the brain. As there is always some BOLD signal in every part of the brain, there is always blood in every part of the body— hopefully, every single entry in the correlation matrices has some weight. We did not need all of these weights; many of them were extraneous, so the weights had to go through some thresholding process. For analytical convenience, we also binarized the matrices as a part of this process.

There are several types of thresholding common in connectomics research. We chose a global thresholding approach. To do this, the dataset was split into a patient group of individuals with Parkinson’s disease and a control group. Then, all of the weights in all of the matrices in these two groups were added to two respective master lists. From these lists, two normal distributions were constructed. For each weight in each graph in each group, a z -test was conducted at an alpha level of 0.05 using the relevant group’s weight distribution. If the p -value was less than alpha, that weight was changed to 1 in the correlation matrix. If the p -value was greater than alpha, the correlation was deemed not statistically significant, and that weight was changed to 0. The distributions of the weights in patient and control groups are shown in figure 3.

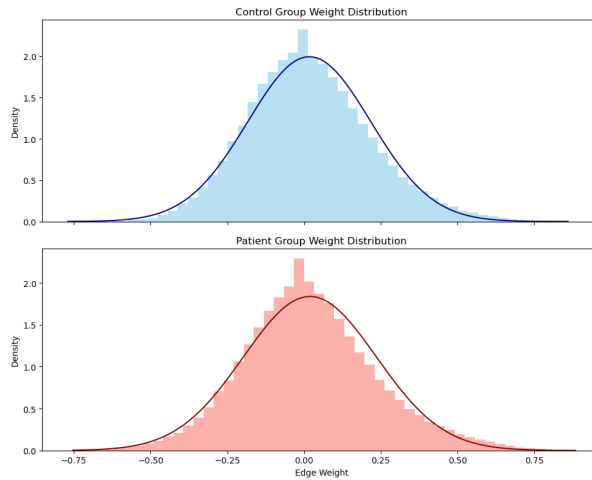


Figure 3: Distribution of Weights in the Patient and Control Groups

This process was done in the DataProcessing Python module mentioned before.

Graph Construction

Once the matrices had been thresholded and binarized, they were treated as adjacency matrices from which graphs were constructed. These graphs, and their visual representations, were constructed in a Python module named GraphConstructor. This module especially leveraged NetworkX. A sample visualization of these graphs can be seen in figure 4.

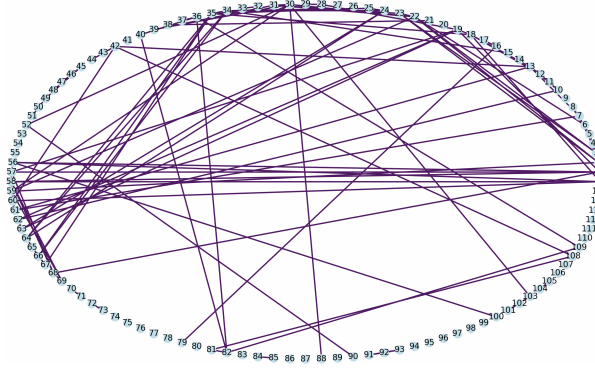


Figure 4: Example brain graph

Analysis

The first step of the analysis of these graphs was to calculate some basic statistics about them. In this phase the number of nodes, numbers of edges, density, and average degree was calculated for each graph, and it was determined that the adjacency matrices were symmetric. An example degree distribution of one of the graphs can be seen in figure 5.

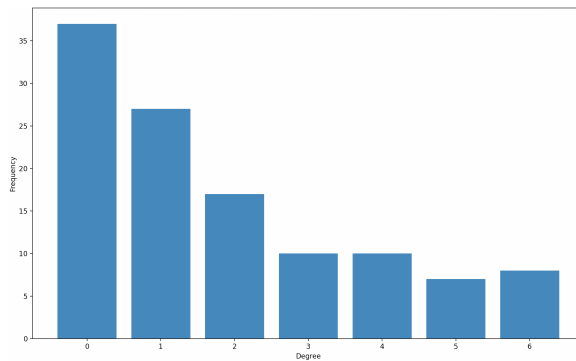


Figure 5: Distribution of node degrees in brain graph

The main analytical interest of this project was to determine the “small-worldness” of the graphs in question. The idea of a small world graph was discussed in the introduction, but no method to

calculate “small-worldness” was given. In fact, Watz and Strogatz provided no such method in their paper. However, since the publication of [2], the idea of a small-world graph has become quite well known, and several papers have proposed methods to calculate “small-worldness”. One such paper is “Network ‘small-world-ness’: A quantitative method for determining canonical network equivalence” by Humphries and Gurney [6]. In this paper, the authors propose a method to calculate the small-world coefficient (SWC). This is done by calculating the CC and CPL of the graph in question and comparing these values to the CC and CPL of a randomly generated Erdos-Renyi graph with the same number of nodes and edges. An Erdos-Renyi graph is a graph constructed by determining a specific number of nodes, 116 in our case, and then applying a probability to every possible pair of nodes that there will be an edge connecting them. Our comparison between the real and Erdos-Renyi graphs is done through the following equation, where CC_{real} and CPL_{real} are the CC and CPL of the original graph and CC_{rand} and CPL_{rand} are the same for the randomly generated graph.

$$SWC = \frac{\frac{CC_{real}}{CC_{rand}}}{\frac{CPL_{real}}{CPL_{rand}}} \quad (1)$$

To calculate the CC of a graph, we use,

$$CC = \frac{3t}{p_2} \quad (2)$$

where t is the number of triangles in the graph, and p_2 is the number of paths of length 2 in the graph.

To calculate the CPL of a graph we use,

$$CPL = \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^n \frac{g_{ij}}{n-1} \right) \quad (3)$$

where n is the number of nodes in the graph and g_{ij} is the shortest possible path between nodes i and j .

Equation (2) is relatively straightforward and poses no problems for our purposes. However, equation (3) is unfortunately problematic. Because our graphs have gone through a thresholding process to reduce the influence of noise in the data, there is no reason that they have to be connected. The graph in figure 3 is not connected, for example. This makes the calculation of g_{ij} impossible for some pairs of nodes, because there is no path between them at all.

We resolve this issue by modifying equation (3) slightly to get,

$$CPL = \frac{1}{n_S} \sum_{i=1}^{n_S} \left(\sum_{j=1}^{n_S} \frac{g_{ij}}{n_S - 1} \right) \quad (4)$$

where n_s is the number of nodes in the largest connected subgraph of the original graph, and i and j are nodes in that subgraph.

The random Erdos-Renyi graphs for calculating CPL_{rand} and CC_{rand} graphs were generated using NetworkX, specifically the `gnm_random_graph` function. An example of one of these random graphs can be seen in figure 6.

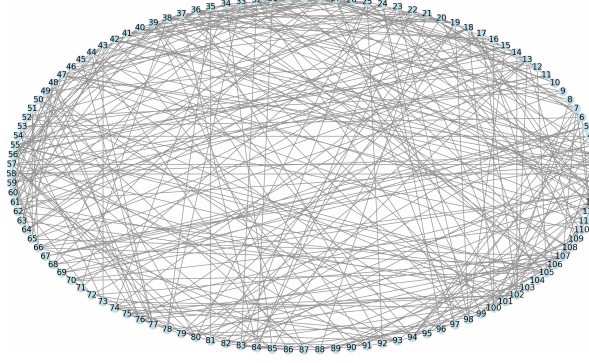


Figure 6: Randomly generated Erdos-Renyi graph based on brain graph

Using these graphs and equations (1),(2), and (4), the SWC of each brain graph was calculated. The relevant statistics pertaining to each graph were then added to a new dataset containing that information for all of the graphs in the dataset. All of the analysis was done in Python in a module named GraphAnalysis.

Results

The methods outlined in the analysis section were used to find statistics concerning individual graphs. The full dataset containing the results for each graph cannot be included here, but the mean values for all of the numeric fields in the results dataset are included in table 1.

Discussion

The results from table 1 seem to confirm our original hypothesis. For the control group, the mean SWC is 2.128, which is higher than that of the patient group at 1.8709. Based on similar studies, like [7], we expected to find this.

Having confirmed that the SWC of the control group is indeed higher than that of the patient group, we are now faced with a new question: why? The biggest disparities in the results from table 1 are in the edges, density, and average degree, where the patient group has significantly higher values. Thus, the patient group is significantly more connected than the control group. This would seemingly undercut our

Table 1: Mean stats from control and patient groups

Statistic	Mean (Patients)	Mean (Controls)
nodes	116	116
edges	402.44	258.929
possible_edges	6670	6670
density	0.0603	0.0388
average_degree	6.9386	4.4643
CCrand	0.1383	0.0963
CCreal	0.3443	0.306
CPLrand	2.2583	2.3631
CPLreal	3.5332	3.901
small_world_coefficient	1.8709	2.128
number_of_subgraphs	23.92	27.5714
biggest_subgraph_nodes	86.32	81.0714
biggest_subgraph_edges	394.32	248.929

result concerning the SWC. At first glance, one would expect the 'small-worldness' of a graph to increase as the number of edges. As more edges are added to a graph, the CC should increase— more edges means more clustering. Similarly, as the number of edges increases, more paths are made available, so the CPL should go down. However, Figure 7 shows that SWC decreases as the number of edges increases.

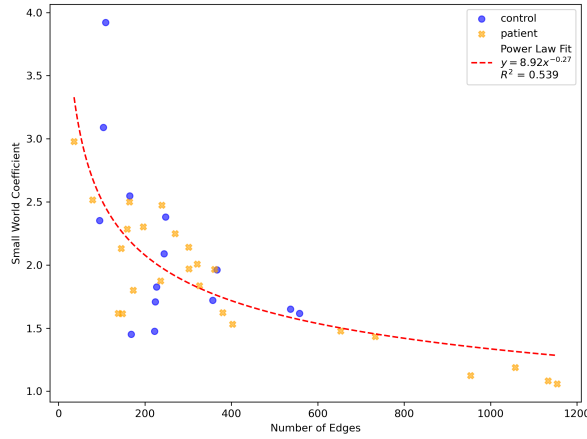
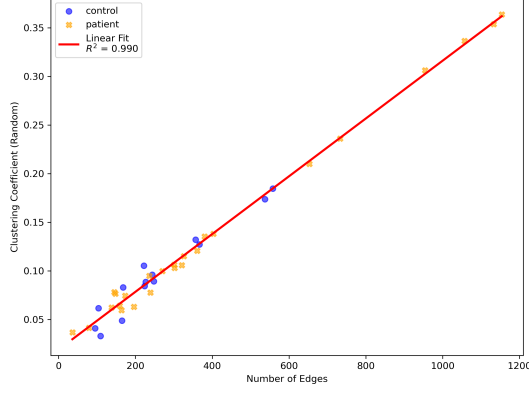
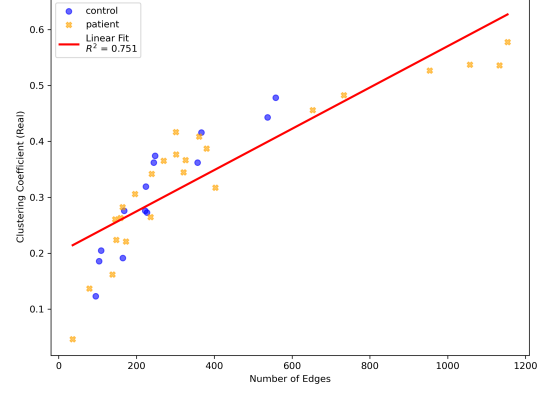


Figure 7: Small World Coefficient vs. Edges

To understand why the SWC decreases rather than increases as the number of edges increases, we must recall the method for calculating the SWC. Looking at equation (1) we see that the SWC increasing is not simply a matter of the CC increasing and the CPL decreasing. They must both do this faster than their random counterparts. Figure 7 clearly indicates that one or both of these metrics is not doing this.



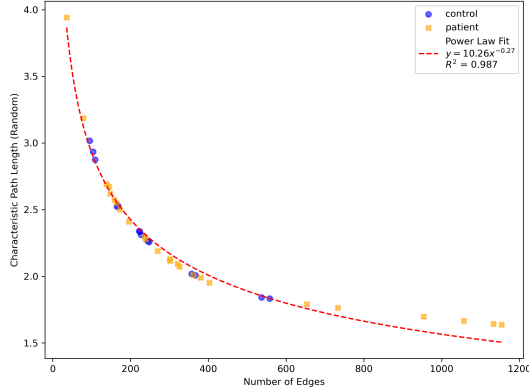
(a) Random Graph CC vs Edges



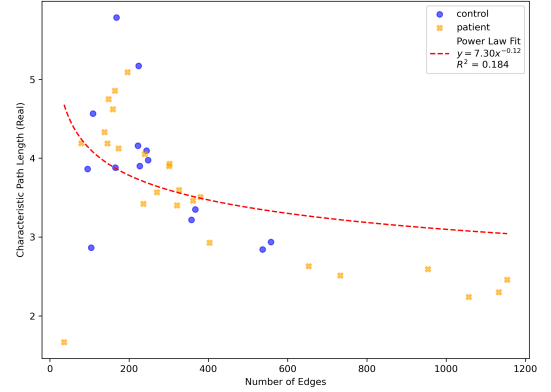
(b) Real Graph CC vs Edges

Figure 8: Random and Real Clustering Efficients vs Numbers of Edges

Figure 8 shows us that the CC of the brain graphs initially increases more rapidly than the CC of the random graphs as the number of edges increases, and, although its rate of increase levels off, it starts and remains at a higher value than the CC of the random graphs. Hence, we can say that the CCs of the brain graphs have values that are outperforming those of the random graphs as the number of edges increases.



(a) Random Graph CPL vs Edges



(b) Real Graph CPL vs Edges

Figure 9: Random and Real Characteristic Path Length vs number of Edges

Figure 9 is telling. As the number of edges increases, the values for CPL for the brain graphs do not decrease as fast as the values for the CPL of the random graphs. Hence, the CPL for the brain graphs does not decrease faster than its random counterpart.

We have determined that the CC of the brain graphs increases faster than that of the random graphs as the number of edges increases, but the CPL of the brain graphs decreases slower than that of the random graphs as the number of edges increases.

Looking at equation (1), it must necessarily be the case that the relatively slow rate of decrease of the

CPL of the brain graphs as the number of edges increases must have a greater effect than the relatively fast rate of increase of the CC as the number of edges increases. In simpler words, the denominator needs to increase faster than the numerator for the value of the fraction to go down. This is the only way the SWC could decrease as the number of edges increases in our case.

In order for this to account for our finding that the average SWC of the control group is greater than that of the patient group, one more thing must be true: the graphs in the patient group must have more nodes. This can be seen indirectly in figures 7, 8, and 9, but it is shown definitively in figure 10.

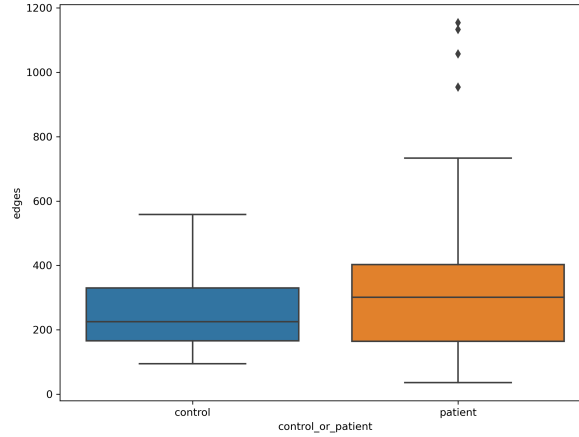


Figure 10: Box Plot of Edges by Group

A corollary question from this line of reasoning is: why do the graphs in the patient group have more edges? The reason for this must have something to do with the thresholding process. Each graph in both groups had the same amount of edges before thresholding and a different amount after. For some reason, more edges passed the z -tests in the patient group than in the control group. The differences in the distributions of weights in each group, as shown in figure 3, might be able to shed some light on this. The patient distribution being narrower might permit more values to pass the z -test. Perhaps entirely different thresholding methods could be used to avoid this issue. However, that is beyond the scope of this paper. More research is needed in these areas.

Conclusion

This study explored the network structure of brain graphs in Parkinson’s patients and healthy controls using macroscale connectomic methods. Small-world coefficients are greater in the control group. Additionally, variations in graph density and node degree suggest potential differences in network organization. These results: suggest that Parkinson’s disease could cause a degradation in small world structure in functional brain graphs, highlight the complexity of interpreting functional connectivity, and point to the

need for deeper analysis of thresholding methods and BOLD signal variability in neurological disorders like Parkinson's.

References

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